it for research on supply chain issues and oil pipelines, leveraging an LLM to identify potential problems. Additionally, this approach is commonly deployed for fraud detection use cases.

Knowledge graphs are the industry standard for fraud detection, but the range of scenarios where knowledge graphs + LLMs can provide significant advantages is vast. This includes areas like cybersecurity, banking, manufacturing, and supply chain management.

Q. How is the market in India for your solution?

A. In India, we have numerous customers ranging from consulting firms and banks to technology companies. During the presentation I gave at Open Source India 2024, many attendees were familiar with Neo4j or are using it for internal company projects. Most financial companies are already using Neo4j, particularly for fraud detection use cases.

Developers are increasingly adopting Neo4j for various application development use cases because it is much more intuitive to work with knowledge graphs and LLMs compared to other types of databases, which are not as well-suited for LLM use cases.

Q. How equipped are knowledge graphs to be scalable and flexible for different customer requirements?

A. I think knowledge graphs are very broad and scalable, making them suitable for a variety of use cases. At its core, a knowledge graph is simply a collection of nodes and relationships, with the ability to run queries. We also have data science extensions built on top of it. However, beyond that, you can create very powerful and specific applications for a wide range of industries.

It's like how relational databases have become the core for most time-based or tabular data, which is organised in rows and columns. Graph databases, on the other hand, are a natural format for anything that involves networks or relationship-based document structures, which are closely aligned with what you need when building AI/ML applications and working with generative AI technology.

Q. How does the knowledge graph know that this is the relevant information it needs to extract?

A. There are many effective algorithms in knowledge graphs for evaluating closeness and the proximity of different nodes, as well as their relationships. With a well-constructed knowledge graph, you can efficiently perform multi-level hops around the graph to retrieve relevant information that is contextually useful for an LLM or an application to generate results. The same structure in tables and joins would be difficult to construct and would also require dozens of table joins to achieve the same results, leading to very slow performance. In contrast, this is a natural representation for knowledge graphs and graph-based applications.

Q. Can GraphRAG be applied to other sector domains like data analytics and business intelligence, apart from GenAI?

A. Yes. Graph databases have a wide range of applications and use cases beyond GenAI and LMs. For instance, Neo4j's data science package enables users to gain insights and perform in-depth analysis of their graph data.

Many of our customers use graph databases for fraud prevention and supply chain optimisation, among other purposes. In fact, graph databases are often the optimal choice for these kinds of applications. Historically, graph databases were foundational for building recommendation engines, and they remain widely used for that purpose today.

Every time you search on Google or a major search engine, a graph database is working behind the scenes to deliver results. This is because algorithms like PageRank and other recommendation engine technologies were originally developed as graph science applications.

Q. Are there any metrics to measure if the solution obtained with a combination of a knowledge graph and LLM is the best or the most effective solution?

A. Measuring the responses from LLMs can be challenging. Knowledge graphs, however, offer the advantage of providing specific nodes and portions of the graph that are passed into the model, making the process less of a black box. This allows you to analyse the results, identify the data that was passed in, and evaluate how effectively the model delivers accurate responses.

Recent research papers have demonstrated that the combination of knowledge graphs and LLMs offers a lot of benefits. This approach not only enhances research outcomes but also proves advantageous across various industries.

Q. What does the future look like for knowledge graphs?

A. This year has been marked by the rise of generative AI and RAG, but moving forward, GraphRAG and knowledge graphs will represent the next evolution, delivering improved accuracy and greater explainability in the results. Neo4j stands out as the best open source and commercial graph database for addressing these use cases.

Our goal is to empower the developer community to effectively leverage generative AI combined with LLMs and graph use cases. We are committed to providing valuable content, as well as offering more hands-on learning opportunities. END